



Harrison Wekesa Wanjala

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Experienced ICT professional from Kenya skilled in networking, programming, and IoT (Arduino, Python, C++). Expertise in automated MikroTik internet solutions, school digital systems development, ICT course content creation, entrepreneurship research, and founder of Bamboo Technologies merging technology, business, and education.

Professional Experience

Kitale National Polytechnic

01/2022 - 05/2025

ICT Technician

- Prepare and maintain ICT labs, ensuring functionality of computers, projectors, and networking equipment for lessons
- Install and configure teaching software for practical sessions
- Troubleshoot technical issues to minimize disruptions during lessons
- Support trainers in networking, programming, IoT, and other ICT topics
- Set up and manage lab networks, virtual environments, and IoT kits
- Configure and secure practical assessment environments
- Maintain and update digital learning materials and software tools for class
- Assist in creating multimedia content for teaching
- Manage e-learning portals for student access to resources
- Guide responsible use of lab equipment and ICT resources for students
- Provide basic troubleshooting for student devices
- Support students with learning platform accounts and assessments
- Introduce new software and emerging technologies in class
- Incorporate cybersecurity practices into student projects
- Promote safe and ethical ICT use in class activities
- Manage digital systems for assessments
- Ensure secure exam environments and prevent unauthorized access
- Provide technical assistance during e-assessments and online submissions

Kitale National Polytechnic

06/2022 - present

ICT Trainer

- Deliver theory and practical lessons in networking, programming, IoT, and related ICT topics
- Develop TVET-curricula-aligned lesson plans emphasizing industry standards
- Integrate emerging technologies and industry practices into training
- Guide students in hands-on network setup, coding, and IoT projects
- Demonstrate software installation, configuration, and troubleshooting
- Supervise lab sessions for ICT equipment safety and proper use
- Create teaching materials, lab manuals, and multimedia resources
- Update digital learning platforms with notes, assignments, and tutorials
- Integrate e-learning tools and simulations for interactive learning
- Provide academic guidance and technical assistance to students
- Mentor students on ICT career opportunities and industry trends
- Encourage innovation and teamwork in problem-solving projects
- Administer written and practical assessments, ensuring compliance

- Evaluate student performance, provide feedback, and record results
 - Incorporate new tools, programming frameworks, and IoT solutions
 - Support students in digital research, project development, and presentations
 - Promote safe, ethical, and professional ICT use in training
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Education

P.A.G Maina Primary School KCPE - 395 • 395 Marks	01/2010 - 12/2013
Friends School Kamusinga KCSE - B • KCSE B	01/2014 - 12/2017
Kabarak University BSc. Information Technology - Second class upper division • Second class upper division	05/2018 - 12/2022
Kabarak University MSc. Information Technology - Ongoing • Completed first year studies.	01/2023 - present

Projects

Automated MikroTik Hotspot Voucher System • Configured MikroTik RouterBOARD to integrate with M-Pesa for automated voucher generation and internet access control.	01/2025 - 05/2025
School Digital Workflow Automation • Designed a Google Forms + Google Sheets system for managing over 5,000 students, 50+ courses, and multiple assessments in a TVET institution.	06/2025 - 06/2025
Entrepreneurship & Innovation Research Paper • Drafted and refined content for Harnessing TVET Innovation Hubs for Economic Inclusion of Youth and Women.	05/2025 - present

Key Skills

- Networking (LAN/WAN/Wi-Fi configuration)
- MikroTik RouterOS setup & hotspot management
- Programming (Python, C++, Arduino IDE)
- IoT development & integration
- ICT lab setup, maintenance, troubleshooting
- Google Workspace & Microsoft 365 administration
- Digital workflow automation (Google Forms + Sheets)
- E-learning platforms, online assessment tools
- Data backup, recovery, cybersecurity
- Software installation, configuration, support

- Lesson planning, competency-based training
 - Practical assessment preparation, supervision
 - Research, innovation project support
 - ICT asset management, inventory control
 - Content creation for tech education
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Interests

- Emerging technologies in ICT and IoT
 - Networking and internet optimization
 - Programming and software development
 - ICT in education and skills
 - Digital transformation in TVET
 - Cybersecurity awareness
 - Automation and smart systems
 - E-learning innovation
 - Research on empowerment through tech
 - Entrepreneurship and tech development
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References

Eubert Asena - Director, Asena logistics - Nairobi
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